Secure Office Network

Tool: Cisco Packet Tracer

About This Portfolio

This portfolio documents two hands-on cybersecurity projects completed as part of professional development alongside active work experience. These projects demonstrate practical skills in network design, VLAN segmentation, access control, and security log analysis.

What Was Built

- Designed a 3-department office network: ADMIN (VLAN 10), HR (VLAN 20), IT (VLAN 30)
- Configured 3 switches — one per department — with VLAN access ports and trunk uplinks
- Set up a router with sub-interfaces (Router-on-a-Stick) for inter-VLAN routing
- Assigned IP subnets: 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24
- Implemented ACL 101 to block HR department from accessing IT department
- Verified full connectivity and security policy using show commands

Network Design Summary

VLAN 10 — ADMIN	192.168.10.0/24 | Gateway: 192.168.10.1 | Switch Fa0/1, Fa0/2
VLAN 20 — HR	192.168.20.0/24 | Gateway: 192.168.20.1 | Switch Fa0/1, Fa0/2
VLAN 30 — IT	192.168.30.0/24 | Gateway: 192.168.30.1 | Switch Fa0/1, Fa0/2
Trunk Ports	Fa0/3 on each switch — 802.1Q encapsulation
Router Interfaces	Gi0/0.10, Gi0/0.20, Gi0/0.30 (sub-interfaces)
ACL 101	Deny HR (192.168.20.0) → IT (192.168.30.0) | Permit any any